

Distributed Smart Traffic Light Control System

Modern urban traffic systems are often centralized and unable to react efficiently to rapidly changing traffic conditions. This leads to congestion, increased travel time, and higher environmental impact due to unnecessary vehicle idle time.

The motivation behind this project is to explore a decentralized and distributed approach to traffic management, where each intersection can operate independently while still cooperating with neighboring intersections.



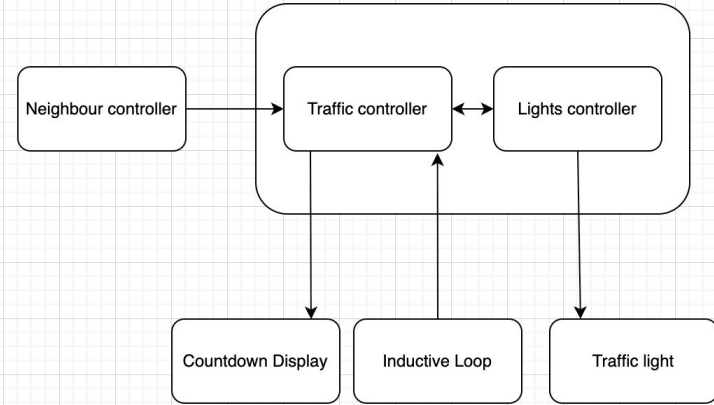
Introduction

1. VSCode
2. Node.js
3. MQTT
4. JSON
5. Grafana
6. Influxdb

Choice of tools

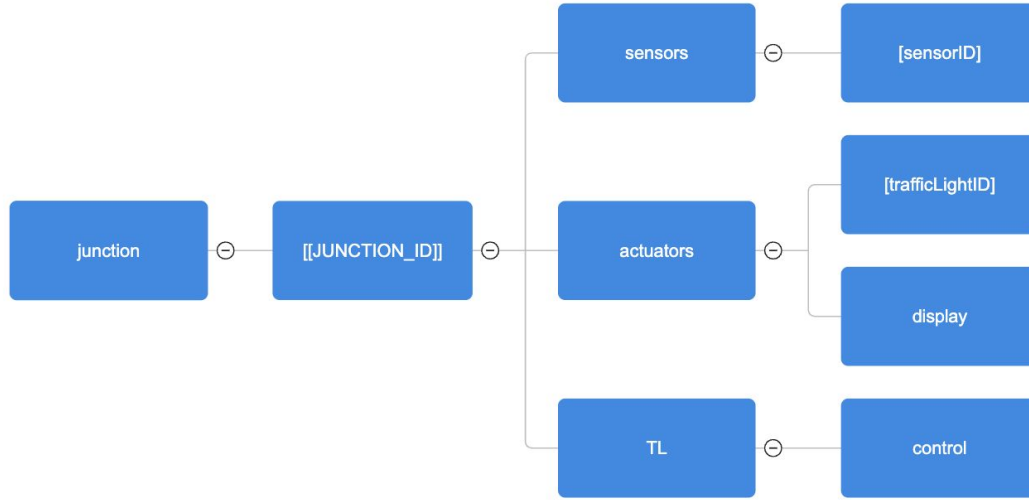
Implemented edge devices

Implemented controllers

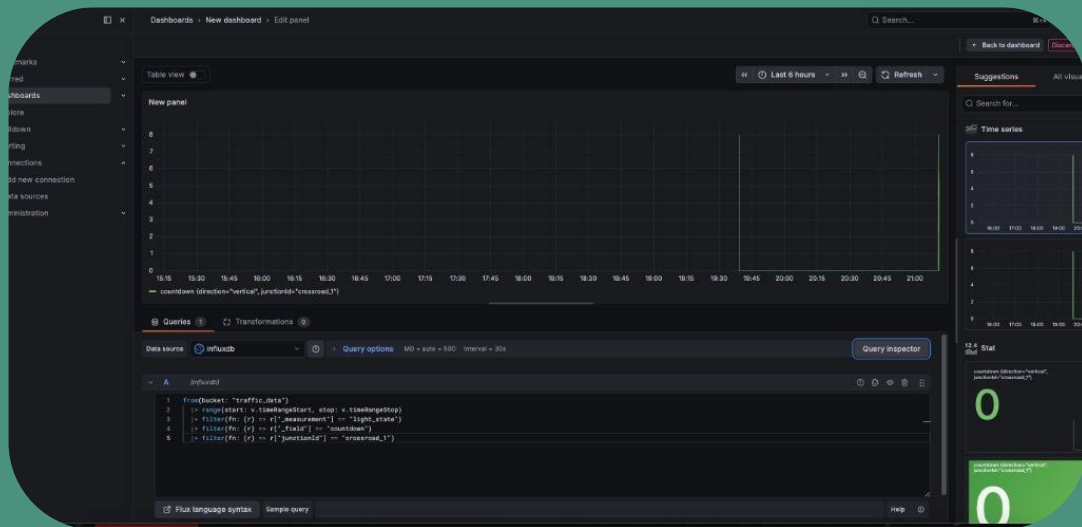


Components communication

(all via mqtt)



MQTT topics



Grafana

Live demo